Indices, Surds & Quadratics

ANSWER KEY

Section 1: Laws of Indices & Surd Arithmetic

1 Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$ ✓

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

Apply the power $1/3$ to each term: $8^{1/3}=2$, $(x^6)^{1/3}=x^2$, $27^{1/3}=3$, $(y^3)^{1/3}=y$. Result is $\frac{2x^2}{3y}$.

2 Simplify $\sqrt{48}-\sqrt{12}+\sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$ ✓

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

$\sqrt{48}=4\sqrt{3}$, $\sqrt{12}=2\sqrt{3}$, $\sqrt{75}=5\sqrt{3}$. Then $4\sqrt{3}-2\sqrt{3}+5\sqrt{3}=7\sqrt{3}$.

3 Rationalize the denominator of $\frac{6}{3-\sqrt{3}}$ and write in simplest form.

a. $3+\sqrt{3}$ ✓

b. $2+\sqrt{3}$

c. $3-\sqrt{3}$

d. $6+2\sqrt{3}$

Multiply numerator and denominator by $(3+\sqrt{3})$: $\frac{6(3+\sqrt{3})}{9-3}=\frac{6(3+\sqrt{3})}{6}=3+\sqrt{3}$.

Section 2: Completing the Square & Turning Points

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4 Express $2x^2 - 12x + 11$ in the form $a(x-h)^2 + k$.

a. $2(x-3)^2 - 7$ ✓

b. $2(x-3)^2 + 11$

c. $2(x-6)^2 - 25$

d. $2(x-3)^2 - 11$

$2(x^2 - 6x) + 11 = 2[(x-3)^2 - 9] + 11 = 2(x-3)^2 - 18 + 11 = 2(x-3)^2 - 7$.

5 Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. **Minimum is -7 at $x = -3$** ✓

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

$3(x^2 + 6x) + 20 = 3[(x+3)^2 - 9] + 20 = 3(x+3)^2 - 27 + 20 = 3(x+3)^2 - 7$. Minimum is -7 when $x = -3$.

6 The curve $y = -(x+4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. **Maximum at $(-4, 15)$** ✓

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

Since the leading coefficient is negative, the parabola opens downward, giving a maximum at $(-4, 15)$.

Section 3: The Discriminant & Nature of Roots

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7 Find the set of values of k for which the equation $x^2 + kx + (k+3) = 0$ has two distinct real roots.

a. $k < -2$ **or** $k > 6$ ✓

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

Distinct real roots require $\Delta > 0$: $k^2 - 4(1)(k+3) > 0 \Rightarrow k^2 - 4k - 12 > 0 \Rightarrow (k-6)(k+2) > 0$. Thus $k < -2$ or $k > 6$.

8 The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2 ✓

b. 2

c. 7

d. -6

Set $x^2 - 4x + 7 = 2x + c \Rightarrow x^2 - 6x + (7 - c) = 0$. For tangency, $\Delta = 0$:
$36 - 4(7 - c) = 0 \Rightarrow 36 - 28 + 4c = 0 \Rightarrow 4c = -8 \Rightarrow c = -2$.

9 Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$ ✓

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

No real roots means $\Delta < 0$: $16 - 4p^2 < 0 \Rightarrow 4p^2 > 16 \Rightarrow p^2 > 4$. Since $p > 0$, $p > 2$.

Section 4: Disguised Quadratics & Exam-Style Modeling

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10 Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$ ✓

b. $x = \pm 2$

c. $x = \pm 3,\ \pm 2i$

d. $x = 9$

Let $u = x^2$: $u^2 - 5u - 36 = (u - 9)(u + 4) = 0$. Since x is real, $u = x^2 = 9 \Rightarrow x = \pm 3$ ($u = -4$ has no real solutions).

11 Solve the equation $2^{2x} - 10(2^x) + 16 = 0$.

a. $x = 1$ **or** $x = 3$ ✓

b. $x = 2$ or $x = 8$

c. $x = 0$ or $x = 4$

d. $x = -1$ or $x = 3$

Let $u = 2^x$: $u^2 - 10u + 16 = 0 \Rightarrow (u - 2)(u - 8) = 0$. Thus $2^x = 2 \Rightarrow x = 1$, and $2^x = 8 \Rightarrow x = 3$.

12 A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ \text{m}^2$ ✓

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

Let width be x. Length along wall is $80 - 2x$. Area
$A(x) = x(80 - 2x) = 80x - 2x^2 = -2(x - 20)^2 + 800$. The maximum area is $800\ \text{m}^2$ at $x = 20$ m.

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